FUNCTIONS

Chapter 7

Polynomials

7

SYLLABUS 2.12 HL only

Nature rarely changes in sudden jumps. Temperature, water level and the position of a moving object all change smoothly. Polynomials are the simplest functions that behave the same way, so they are used to model a great deal of the physical world.

The degree matches the situation. Distance travelled at a constant speed is linear. A thrown ball follows a parabola, the graph of a quadratic, and so do the surface of spun water and the cross-section of a satellite dish. The volume of a sphere grows with the cube of its radius, $V = \frac{4}{3}\pi r^3$, and the power a ship needs grows roughly with the cube of its speed. An evenly loaded beam bends into a quartic curve. Over a short enough interval, almost any smooth curve can be matched closely by a polynomial, so measurements are often fitted with one when no exact formula is known. See Ch. 17.

That smoothness comes from the way a polynomial is built. A polynomial uses one variable and a few constants, combined by addition and multiplication only. There is no division by the variable, no square root, and no trigonometry. A polynomial is therefore defined for every real number, so its graph has no gaps and no asymptotes, and it has no corners anywhere. The sum of two polynomials is a polynomial, and so is their product.

Describing a polynomial is easy. Solving one is not. To solve $f(x) = 0$ is to find every value of x that makes the polynomial equal to zero. An exact formula for those values exists for degree 2, degree 3 and degree 4. For degree 5 and above, no such formula exists. Niels Abel proved this in 1824. This chapter gives the methods that work without a formula: testing whether a value is a root, factorising a polynomial once one root is known, and finding the sum and product of the roots from the coefficients alone.

Before any of that you will learn to read a polynomial's graph from its factorised form: where it crosses the axis, where it only touches, and what it does at either end. That is the thread running through the chapter. A polynomial's coefficients and its roots describe the same object, and nearly every technique here moves between the two.

7.1 Polynomial Functions and their Graphs

Much of the graph of a polynomial can be described before a single point is plotted. The degree and the coefficients carry that information, so begin with the formal definition.

DEF A **polynomial function** of **degree** n has the form

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0, \qquad a_n \neq 0,$$

where the coefficients are constants and n is a non-negative integer.

Polynomials are named by their degree. Degree 1 is **linear**, degree 2 is **quadratic**, degree 3 is **cubic**, degree 4 is **quartic**, and degree 5 is **quintic**. The degree also limits the shape: it limits how many times the curve can meet the x-axis, and how often it can change direction.

The constants $a_n, a_{n-1}, \dots, a_1, a_0$ are the **coefficients** of the powers of x. The coefficient of the highest power is a_n, called the **lead coefficient** (also written leading coefficient), and the term $a_n x^n$ that contains it is the **leading order term**.

A coefficient may take any value, with one restriction: the lead coefficient cannot be zero. If a_n were zero, the term $a_n x^n$ would be zero, and what remained would be a polynomial of degree $n-1$ instead.

The sign of the lead coefficient determines which way the ends of the curve point. A polynomial with $a_n > 0$ is sometimes called a **positive polynomial**, and one with $a_n < 0$ a **negative polynomial**. These names describe the lead coefficient only. They do not mean that the function is positive or negative everywhere: $y = x^2 - 4$ has a positive lead coefficient, yet y is negative for $-2 < x < 2$.

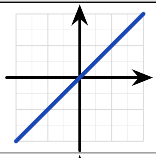
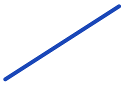
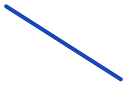
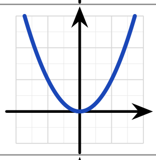
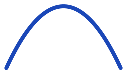
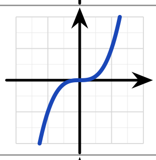
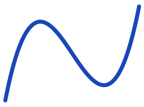
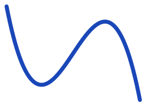
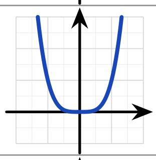
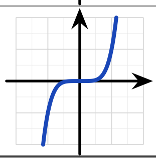
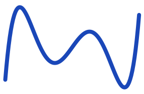
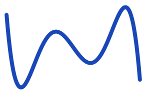
Three features are visible before you plot a single point.

The **end behaviour**, meaning the direction of the curve as $x \to \pm\infty$, is determined entirely by the leading term. If n is odd, the two ends point in opposite directions. If n is even, they point the same way, and the sign of a_n determines whether that direction is upward or downward.

The number of **roots**, the places where the graph meets the x-axis, can never exceed the degree. A cubic meets the axis at most three times, a quartic at most four.

A **turning point** is a place where the curve stops rising and begins to fall, or stops falling and begins to rise. A polynomial of degree n has at most $n-1$ turning points. The possible numbers then decrease in steps of two: $n-1$, $n-3$, and so on. The number changes by two rather than by one because a maximum and a minimum are always lost together, never one alone.

The table collects all three features for the degrees you meet most often.

n	$y = x^n$	**Shape when** $a_n > 0$	**Shape when** $a_n < 0$	**Times it meets the** x**-axis**	**Turning points**
1				1	0
2				0, 1 or 2	1
3				1, 2 or 3	0 or 2
4				0, 1, 2, 3 or 4	1 or 3
5				1, 2, 3, 4 or 5	0, 2 or 4

Where a factor is repeated, the graph behaves differently. The **multiplicity** of a root is the number of times its factor appears in the factorised polynomial. In $(x-1)(x-2)^2$ the root $x = 1$ has multiplicity one, and the root $x = 2$ has multiplicity two.

Multiplicity decides what the curve does at the axis. A root of multiplicity one, from a single factor $(x-a)$, crosses the axis. A root of even multiplicity, such as $(x-a)^2$, only *touches* the axis and turns back. Practise both directions: reading the multiplicities from a sketch, and predicting the sketch from the factors.

EXAMPLE 7.1

Sketch $y = (x+1)(x-2)^2$, showing the intercepts and the behaviour at each root.

Read the roots and their multiplicities from the factors. The factor $(x+1)$ is single, so the graph crosses the axis at $x = -1$. The factor $(x-2)^2$ is squared, so the graph touches the axis at $x = 2$ and turns back.

Find the y-intercept. Substitute $x = 0$: $y = (1)(-2)^2 = 4$, the point $(0, 4)$.

Fix the ends from the leading term. Multiplying out leads with x^3: an odd degree with a positive coefficient, so the curve runs from $-\infty$ at the left to $+\infty$ at the right.

These three facts fix the shape.

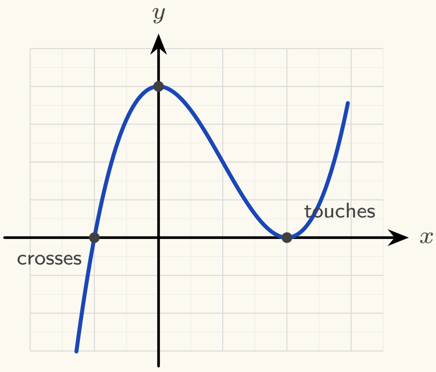


Figure 7.1 A sketch of $y = (x+1)(x-2)^2$. The simple root at $x = -1$ crosses the axis; the double root at $x = 2$ touches it and turns back. The y-intercept is $(0, 4)$, and the ends follow x^3.

Every factorised polynomial is sketched the same way: the roots and their multiplicities give the behaviour at the x-axis, the value at $x = 0$ gives the y-intercept, and the leading term gives the two ends.

When the polynomial is written out in full rather than factorised, the y-intercept is even quicker to read. Every term except the constant contains x, so substituting $x = 0$ makes all of them zero. The constant term a_0 is what remains.

KEY The graph of $y = a_n x^n + \cdots + a_1 x + a_0$ crosses the y-axis at $(0, a_0)$. *The constant term is the y-intercept.*

For $y = 2x^3 - 3x + 5$, the constant term is 5, so the graph crosses the y-axis at $(0, 5)$.

From the graph back to the equation

The reverse problem is also examined. Given a sketch, you are asked for the polynomial that produced it. The intercepts give you almost everything you need.

Each x-intercept contributes a factor, and the behaviour there gives its multiplicity: a crossing gives a single factor, a touch gives a squared factor. That fixes the polynomial up to one unknown number, the leading coefficient, and any other point on the curve determines it.

KEY **Finding a polynomial from its graph.** Write one factor for each x-intercept, with the multiplicity the graph shows at that point. Then substitute the coordinates of any other known point to find the leading coefficient a_n.

EXAMPLE 7.2

The graph below shows a cubic. It touches the x-axis at $x = -1$, crosses it at $x = 2$, and passes through $(0, -1)$. Find its equation.

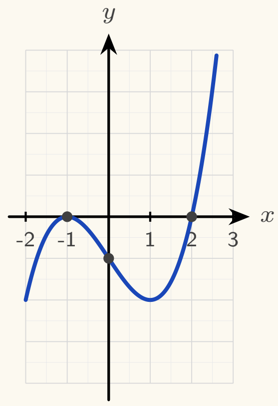


Figure 7.2 The curve touches the x-axis at $x = -1$, crosses it at $x = 2$, and passes through $(0, -1)$. These three facts determine the equation.

Write the factors from the x-intercepts. The curve touches at $x = -1$, so that root is repeated and contributes $(x + 1)^2$. It crosses at $x = 2$, so that root is single and contributes $(x - 2)$. Those factors already account for degree three, so

$$y = a(x + 1)^2(x - 2)$$

for some constant a.

Find a from the remaining point. Substitute $x = 0$ and $y = -1$:

$$-1 = a(1)^2(-2) = -2a,$$

so $a = \frac{1}{2}$ and

$$y = \tfrac{1}{2}(x + 1)^2(x - 2).$$

Check the answer against the graph. The leading term is $\frac{1}{2}x^3$, which is positive, so the curve rises to the right, and the sketch agrees.

The intercepts fix the factors; one further point fixes the leading coefficient.

YOUR TURN 7.1 Polynomial Functions and their Graphs

1. State the degree and the lead coefficient of each polynomial.

(a) $f(x) = 3x^4 - 2x + 7$ [2] (b) $f(x) = 5 - x^3$ [2]

(c) $f(x) = (x - 1)(x + 2)(x - 3)$ [2] (d) $f(x) = -2x^5 + x^2$ [2]

2. Describe the end behaviour of each graph as $x \to +\infty$ and as $x \to -\infty$.

(a) $y = x^3$ [2] (b) $y = -x^4$ [2]

(c) $y = 2x^5 - x$ [2] (d) $y = 3 - x^2$ [2]

3. Answer each question about degree and shape.

(a) State the maximum number of real roots of a degree-5 polynomial. [1]

(b) State the maximum number of turning points of a quartic. [1]

(c) List every possible number of turning points a cubic can have. [2]

(d) Explain why a cubic must have at least one real root. [2]

4. State the roots of each polynomial, and say at each one whether the graph crosses or touches the x-axis.

(a) $y = (x-2)(x+3)(x-5)$ [2]

(b) $y = (x-1)^2(x+4)$ [2]

(c) $y = x(x-3)^2$ [2]

(d) $y = (x+1)^2(x-2)^2$ [3]

5. Consider $y = (x+1)(x-3)^2$.

(a) State the roots and their multiplicities. [2]

(b) Find the y-intercept. [2]

(c) State the end behaviour. [2]

(d) Sketch the curve. [3]

6. A cubic touches the x-axis at $x = 2$, crosses it at $x = -1$, and passes through $(0, 4)$.

(a) Write down the factors, with the correct multiplicities. [2]

(b) Find the lead coefficient. [3]

(c) Write the equation of the cubic. [1]

7.2 Polynomial Division

You can already add, subtract and multiply polynomials. Addition and subtraction collect like terms, and multiplication expands every product before collecting. Division is the operation that still needs a method, and it is the one the rest of this chapter depends on.

Factorising a polynomial means writing it as a product. Once you know one factor, the other is found by dividing. So a method for dividing one polynomial by another is needed before any factorising can be done.

Polynomial division follows the pattern of whole-number division. When you divide 47 by 5 you write

$$47 = 5 \times 9 + 2,$$

which records a divisor, a quotient, and a remainder smaller than the divisor. Polynomials behave the same way, with "smaller" measured by degree instead of size.

KEY **The division algorithm.** For polynomials $f(x)$ and $d(x)$, with $d(x)$ not the zero polynomial, there are unique polynomials $q(x)$ and $r(x)$ such that

$$f(x) = d(x)\,q(x) + r(x),$$

where the degree of r is less than the degree of d. *Dividend equals divisor times quotient, plus remainder.*

Fix the four names now, because every method below uses them. The polynomial being divided, f, is the **dividend**. The polynomial you divide by, d, is the **divisor**. The result q is the **quotient**, and the part that remains, r, is the **remainder**. When $r = 0$ the division is exact, and d is a factor of f.

Long division works for a divisor of any degree. Three steps repeat until the remainder has lower degree than the divisor.

Divide the leading term of the dividend by the leading term of the divisor. That result is the next term of the quotient. Multiply the whole divisor by that term, then subtract the product from the dividend. Repeat on what is left.

EXAMPLE 7.3

Divide $2x^3 - 3x^2 + x - 5$ by $(x - 2)$.
Set the work out in columns, one column for each power of x.

$$
\begin{array}{r|rrrrl}
 & & 2x^2 & +x & +3 & \text{quotient} \\
\cline{2-5}
x-2 & 2x^3 & -3x^2 & +x & -5 & \text{dividend} \\
 & 2x^3 & -4x^2 & & & \\
\cline{2-3}
 & & x^2 & +x & & \\
 & & x^2 & -2x & & \\
\cline{3-4}
 & & & 3x & -5 & \\
 & & & 3x & -6 & \\
\cline{4-5}
 & & & & 1 & \text{remainder}
\end{array}
$$

The divisor $x - 2$ is written to the left of the bracket, and the dividend $2x^3 - 3x^2 + x - 5$ inside it. The quotient is written on the top line, one term at each stage, and the remainder appears on the bottom line.
Each stage repeats the same three steps: divide, multiply, subtract.
Stage 1. Divide the leading term of the dividend by the leading term of the divisor: $2x^3 \div x = 2x^2$. Write $2x^2$ in the quotient. Multiply the divisor by it: $2x^2(x-2) = 2x^3 - 4x^2$. Subtract that from the dividend, which leaves $x^2 + x$.
Stage 2. Repeat on $x^2 + x$. Divide: $x^2 \div x = x$. Multiply: $x(x - 2) = x^2 - 2x$. Subtract, which leaves $3x - 5$.

Stage 3. Repeat on $3x - 5$. Divide: $3x \div x = 3$. Multiply: $3(x-2) = 3x - 6$. Subtract, which leaves 1.

The degree of 1 is lower than the degree of $x - 2$, so the division stops.

The quotient is $2x^2 + x + 3$ and the remainder is 1:

$$2x^3 - 3x^2 + x - 5 = (x-2)(2x^2 + x + 3) + 1.$$

Always write the answer in this form. Multiplying out the right side is a complete check of the division.

CAUTION Write the dividend in descending powers and insert a zero term for every missing power before dividing. Divide $x^3 - 7x + 6$ as $x^3 + 0x^2 - 7x + 6$. Without that placeholder the columns shift, and every subtraction after it is wrong.

EXAMPLE 7.4

Divide $x^3 - 7x + 6$ by $(x - 1)$.

There is no x^2 term, so write it as $x^3 + 0x^2 - 7x + 6$ and keep a column for it.

$$\begin{array}{r|rrrr}
 & & x^2 & +x & -6 \\ \hline
x-1 & x^3 & +0x^2 & -7x & +6 \\
 & x^3 & -x^2 & & \\ \hline
 & & x^2 & -7x & \\
 & & x^2 & -x & \\ \hline
 & & & -6x & +6 \\
 & & & -6x & +6 \\ \hline
 & & & & 0
\end{array}$$

The remainder is 0, so the division is exact and $(x-1)$ is a factor. The quotient $x^2 + x - 6$ factorises further:

$$x^3 - 7x + 6 = (x-1)(x^2 + x - 6) = (x-1)(x+3)(x-2).$$

A remainder of zero shows that the divisor is a factor. The next two sections are built on that fact.

YOUR TURN 7.2 Polynomial Division

7. Simplify each expression.

(a) $(x^3 + 2x - 5) + (3x^3 - x^2 + 1)$ [2] (b) $(x^4 + 3x^2 - 1) - (2x^3 - x^2 + 2)$ [2]

(c) $(x^2+2x-1)(x^2-x+3)$ [3] (d) $(x-2)(x^2+2x+4)$ [2]

8. Divide, and state the quotient and the remainder.

(a) x^2+5x+6 by $(x+2)$ [2] (b) x^3-3x^2+2x by $(x-1)$ [3]
(c) x^3+2x^2-x+1 by $(x-1)$ [3] (d) x^3-8 by $(x-2)$ [3]

9. Divide, and state the quotient and the remainder. Insert a zero term for each missing power first.

(a) $2x^3-x^2-3$ by $(x+1)$ [3] (b) x^3+3x^2-4 by $(x+2)$ [3]
(c) $2x^4-3x^3+x-6$ by $(x-2)$ [4]

10. Divide by the quadratic divisor.

(a) x^4-1 by (x^2-1) [3] (b) x^4-1 by (x^2+1) [3]

7.3 The Remainder Theorem

Here are two operations that appear to have nothing to do with each other. *Evaluating* f at 2 gives the height of the graph above one point. *Dividing* f by $(x-2)$ is the long calculation of the previous section. The remainder theorem states that the two produce the same number, and the proof is one line long.

Take the division algorithm with a linear divisor $(x-a)$:

$$f(x) = (x-a)\,q(x) + r,$$

where the remainder r must be a *constant*, because its degree is less than the degree of $x-a$, and less than one means zero.

Now substitute $x = a$ into that identity. The factor $(x-a)$ becomes zero, so the whole first term is zero, leaving $f(a) = r$. The height of the graph at a *is* the remainder, because everything else in the polynomial is divisible by $(x-a)$ and so contributes nothing at that point.

KEY **Remainder theorem.** When a polynomial $f(x)$ is divided by $(x-a)$, the remainder is $f(a)$. *One substitution replaces the entire division.*

EXAMPLE 7.5

Find the remainder when $f(x) = x^3 - 2x + 5$ is divided by $(x - 2)$.
By the remainder theorem, the remainder is

$$f(2) = 2^3 - 2(2) + 5 = 8 - 4 + 5 = 9.$$

No division needed; the answer is a single evaluation.

EXAMPLE 7.6

Find the remainder when $f(x) = 4x^3 - 4x^2 + 5x - 1$ is divided by $(2x - 1)$.
The divisor need not have the form $(x - a)$. The theorem works for any linear divisor, provided you evaluate at the **zero of the divisor**, meaning the value of x that makes the divisor equal to zero. Here $2x - 1 = 0$ when $x = \frac{1}{2}$, so the remainder is

$$f(\tfrac{1}{2}) = 4 \cdot \tfrac{1}{8} - 4 \cdot \tfrac{1}{4} + \tfrac{5}{2} - 1 = \tfrac{1}{2} - 1 + \tfrac{5}{2} - 1 = 1.$$

Evaluate at the zero of the divisor, not at one of its coefficients. Dividing by $(2x-1)$ means substituting $\frac{1}{2}$, never 1 or 2.

EXAMPLE 7.7

The polynomial $f(x) = x^3 + ax + b$ leaves remainder 2 when divided by $(x - 1)$ and remainder -13 when divided by $(x + 2)$. Find a and b.
Each remainder condition is one evaluation, so two conditions give two equations:

$$f(1) = 1 + a + b = 2 \implies a + b = 1,$$

$$f(-2) = -8 - 2a + b = -13 \implies -2a + b = -5.$$

Subtracting, $3a = 6$, so $a = 2$ and $b = -1$. The remainder theorem has turned two divisions into a pair of linear equations; this reversal, from remainders back to coefficients, is the standard examination form of the theorem.

YOUR TURN 7.3 The Remainder Theorem

11. Find the remainder in each case.

(a) $x^3 + 2x - 1$ divided by $(x - 1)$ [2]
(b) $x^3 - 4x^2 + x + 6$ divided by $(x - 2)$ [2]
(c) $2x^3 - 3x + 1$ divided by $(x + 1)$ [2]
(d) $x^3 + x^2 + x + 1$ divided by $(x + 2)$ [2]

12. Find the remainder in each case. The divisor is not of the form $(x-a)$.

(a) $4x^3-4x^2+5x-1$ divided by $(2x-1)$ [3]

(b) $2x^3+x^2-2$ divided by $(2x+1)$ [3]

13. Interpret each statement.

(a) The remainder when $f(x)$ is divided by $(x-3)$ is 0. What does this tell you? [2]

(b) The remainder when $f(x)$ is divided by $(x-2)$ is 7. State $f(2)$. [1]

(c) $f(-1)=0$. Write down a factor of $f(x)$. [1]

(d) x^4-1 is divided by $(x-1)$. Find the remainder without dividing. [2]

14. The polynomial $f(x)=x^3+ax+b$ leaves remainder 2 when divided by $(x-1)$ and remainder -13 when divided by $(x+2)$.

(a) Write down two equations in a and b. [3]

(b) Solve them to find a and b. [2]

7.4 The Factor Theorem

The most useful case of the remainder theorem is when the remainder is zero. If dividing by $(x-a)$ leaves no remainder, then $(x-a)$ divides $f(x)$ exactly: it is a factor. This is what makes factorising possible.

KEY **Factor theorem.** $(x-a)$ is a factor of $f(x)$ if and only if $f(a)=0$.

To factorise an unfamiliar cubic, search for a root by testing small values. The factors of the constant term are the natural candidates. The reason is proved in §7.6: the product of the roots is the constant term divided by the leading coefficient, apart from a sign. So any integer root must divide the constant term. Once one root a is found, $(x-a)$ is a factor, and dividing it out leaves a quadratic you can finish by hand.

EXAMPLE 7.8

Show that $(x-1)$ is a factor of $f(x)=x^3-6x^2+11x-6$, and factorise f completely.

Test $x=1$:

$$f(1)=1-6+11-6=0,$$

so by the factor theorem $(x-1)$ is a factor.

Now divide it out. Instead of long division, *compare coefficients*. The quotient must be a quadratic, so write

$$x^3 - 6x^2 + 11x - 6 = (x-1)(x^2 + bx + c),$$

then compare the coefficients of matching powers on the two sides. Compare the constant terms: $-c = -6$, so $c = 6$. Compare the coefficients of x^2: $b - 1 = -6$, so $b = -5$. The coefficient of x has not been used, so use it as a check: on the right it is $c - b = 6 + 5 = 11$, which matches the left. ✓ Hence

$$f(x) = (x-1)(x^2 - 5x + 6) = (x-1)(x-2)(x-3).$$

The constant term -6 gives the candidates $\pm1, \pm2, \pm3, \pm6$. Testing $x = 1$ succeeded on the first attempt, and comparing coefficients carried out the division with a check included.

TIP Comparing coefficients is usually faster than long division in an examination. Each unknown comes from one comparison, and any coefficient you did not use provides a check. Long division earns the same marks if you prefer it.

YOUR TURN 7.4 The Factor Theorem

15. Decide whether the given linear expression is a factor. Show your working.

(a) Is $(x-2)$ a factor of $x^3 - 3x^2 + 4$? [2]

(b) Is $(x+1)$ a factor of $x^3 + 2x^2 - x - 2$? [2]

(c) Is $(x-1)$ a factor of $x^3 + x^2 + x + 1$? [2]

(d) Is $(x+2)$ a factor of $x^3 + 8$? [2]

16. Factorise each polynomial completely.

(a) $x^3 - x$ [2]

(b) $x^3 - 4x$ [2]

(c) $x^3 + 2x^2 - x - 2$ [3]

(d) $2x^3 + x^2 - 5x + 2$ [4]

17. Answer each part.

(a) One factor of $x^2 - 5x + 6$ is $(x-2)$. State the other. [1]

(b) Show that $x^3 - 3x^2 + 4$ has a repeated factor. [3]

7.5 Solving Polynomial Equations

You already know how to solve linear and quadratic equations. A linear equation is rearranged directly. A quadratic is solved by factorising, by completing the square, or by the quadratic formula.

For a cubic or any higher degree, there is no formula of that kind for you to use. The factor theorem is used instead. Find one root by testing values, divide that factor out, and repeat until what remains is a quadratic. Then solve the quadratic in the usual way.

Every real root is an x-intercept of the graph, so the algebra and the graph always agree.

EXAMPLE 7.9

Solve $x^3 - 3x^2 - x + 3 = 0$.

Test the divisors of 3. At $x = 1$: $1 - 3 - 1 + 3 = 0$, so $(x - 1)$ is a factor. Dividing,

$$x^3 - 3x^2 - x + 3 = (x-1)(x^2 - 2x - 3) = (x-1)(x-3)(x+1).$$

The roots are $x = 1,\ 3,\ -1$.

CAUTION A cubic with real coefficients always has at least one real root, because one end goes to $+\infty$ and the other to $-\infty$, so the curve must cross the axis. If no rational root can be found by testing, one root is irrational and the equation must be solved by technology.

YOUR TURN 7.5 Solving Polynomial Equations

18. Solve each equation. Each is already factorised or factorises at sight.

(a) $(x-1)(x+2)(x-4) = 0$ [1]

(b) $(x-3)^2(x+1) = 0$ [2]

(c) $x^3 = 4x$ [2]

(d) $x^3 + x^2 - 2x = 0$ [2]

19. Solve each equation, using the factor theorem to find the first root.

(a) $x^3 - 6x^2 + 11x - 6 = 0$ [3]

(b) $x^3 - 2x^2 - x + 2 = 0$ [3]

(c) $2x^3 + x^2 - 5x + 2 = 0$ [4]

20. Solve each equation.

(a) $x^4 - 5x^2 + 4 = 0$ *(Hint: substitute $u = x^2$)* [3]

(b) $x^3 - 1 = 0$, giving only the real root, and explain why there is exactly one. [3]

7.6 Sum and Product of Roots

The coefficients of a polynomial determine the sum and the product of its roots. You can therefore find those two values *without solving the equation at all*. This matters whenever the roots are irrational, or when the equation is too hard to solve, because the sum and the product remain easy to obtain.

Take the quadratic case first. If $ax^2+bx+c=0$ has roots α and β, then the polynomial *is* $a(x-\alpha)(x-\beta)$, and expanding tells us what the coefficients must be:

$$a(x-\alpha)(x-\beta)=a(x^2-(\alpha+\beta)x+\alpha\beta)=ax^2-a(\alpha+\beta)\,x+a\,\alpha\beta.$$

Comparing with ax^2+bx+c: the middle coefficient is $b=-a(\alpha+\beta)$ and the constant is $c=a\,\alpha\beta$. Solve each for the roots' totals:

$$\alpha+\beta=-\frac{b}{a},\qquad \alpha\beta=\frac{c}{a}.$$

The minus sign on the sum is not a convention to memorise. It comes from the brackets, one $-\alpha$ and one $-\beta$.

The same expansion explains every degree at once, and the reason is the counting argument from Ch. 2. Expanding

$$a_n(x-r_1)(x-r_2)\cdots(x-r_n)$$

means choosing, from each bracket, either the x or the root, exactly as in the binomial theorem. To build the x^{n-1} term you take x from all brackets but one, so each root appears exactly once, each carrying a single minus sign: the coefficient collects $-(r_1+r_2+\cdots+r_n)$, the *sum*. To build the constant term you take the root from *every* bracket: all n roots multiplied together, carrying n minus signs, which is where the $(-1)^n$ comes from. So the outermost coefficients of a polynomial record the sum and the product of its roots.

> KEY · IB For the equation $a_nx^n+a_{n-1}x^{n-1}+\cdots+a_1x+a_0=0$,
>
> $$\text{sum of roots}=-\frac{a_{n-1}}{a_n},\qquad \text{product of roots}=(-1)^n\,\frac{a_0}{a_n}.$$
>
> *The second coefficient collects the sum of the roots; the constant term collects their product.*

These are the outermost of a whole family of relations (Vieta's formulas), but the sum and product are the two you will use most, for checking answers, for building equations with required roots, and for finding an unknown coefficient.

One more relation fills the gap between them. Building the x^{n-2} term means taking the root from exactly two brackets and x from the rest, so that coefficient collects the roots multiplied together in pairs. For a cubic $a_3x^3+a_2x^2+a_1x+a_0=0$ with roots α,β,γ, the three relations together are

$$\alpha+\beta+\gamma=-\frac{a_2}{a_3}, \qquad \alpha\beta+\beta\gamma+\gamma\alpha=\frac{a_1}{a_3}, \qquad \alpha\beta\gamma=-\frac{a_0}{a_3}.$$

The syllabus asks you to know only the first and last. The middle relation is a short extension, and it is the one to use whenever a question involves all three roots at once.

EXAMPLE 7.10

The equation $2x^2-5x+3=0$ has roots α and β. Find $\alpha+\beta$ and $\alpha\beta$ without solving, then verify. Here $a=2$, $b=-5$, $c=3$, so

$$\alpha+\beta=-\frac{-5}{2}=\frac{5}{2}, \qquad \alpha\beta=\frac{3}{2}.$$

Solving confirms it: the roots are 1 and $\frac{3}{2}$, whose sum is $\frac{5}{2}$ and product is $\frac{3}{2}$. The sum and product were available from the coefficients before the roots were found.

EXAMPLE 7.11

The cubic $x^3-4x^2+x+6=0$ has roots α,β,γ. Write down $\alpha+\beta+\gamma$ and $\alpha\beta\gamma$.
With $a_3=1$, $a_2=-4$, $a_0=6$ and degree $n=3$,

$$\alpha+\beta+\gamma=-\frac{-4}{1}=4, \qquad \alpha\beta\gamma=(-1)^3\,\frac{6}{1}=-6.$$

(The roots happen to be $-1,2,3$: sum 4, product -6, as promised.)

The method is most useful when the roots are irrational. The next three examples never find a root, yet answer precise questions about them. This is a common examination form.

EXAMPLE 7.12

The equation $x^2-3x+1=0$ has roots α and β. Without solving, find $\alpha^2+\beta^2$ and $\dfrac{1}{\alpha}+\dfrac{1}{\beta}$.
The roots are irrational, but their sum and product are not: $\alpha+\beta=3$ and $\alpha\beta=1$. A **symmetric combination** of the roots is an expression whose value does not change when the two roots are exchanged. Both $\alpha^2+\beta^2$ and $\frac{1}{\alpha}+\frac{1}{\beta}$ are symmetric. Every symmetric combination can be rebuilt from the sum and the product. For the sum of squares, start from the square of the sum:

$$(\alpha+\beta)^2=\alpha^2+2\alpha\beta+\beta^2 \implies \alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=9-2=7.$$

For the sum of reciprocals, combine over a common denominator:

$$\frac{1}{\alpha}+\frac{1}{\beta}=\frac{\alpha+\beta}{\alpha\beta}=\frac{3}{1}=3.$$

The identity $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$ is used often in this topic. Learn it as a rearrangement of $(\alpha+\beta)^2$, not as a separate formula.

EXAMPLE 7.13

With α, β as above, find a quadratic equation with integer coefficients whose roots are 2α and 2β.

A quadratic is determined by the sum and product of its roots: $x^2-(\text{sum})x+(\text{product})=0$. The new totals follow from the old ones:

$$2\alpha+2\beta=2(\alpha+\beta)=6, \qquad (2\alpha)(2\beta)=4\alpha\beta=4.$$

So the required equation is

$$x^2-6x+4=0.$$

No root was ever computed: transforming an equation's roots means transforming their sum and product, nothing more.

EXAMPLE 7.14

The roots of $x^2+kx+12=0$ differ by 1. Find the possible values of k.

Let the roots be α and β with $\alpha-\beta=1$. The coefficients give $\alpha+\beta=-k$ and $\alpha\beta=12$. Connect the difference to these totals by squaring it:

$$(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta \implies 1=k^2-48,$$

so $k^2=49$ and $k=\pm 7$. Check with $k=-7$: the equation $x^2-7x+12=0$ has roots 3 and 4, which indeed differ by 1. A condition on the roots became an equation in the coefficient, using the identity $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$.

YOUR TURN 7.6 Sum and Product of Roots

21. State the sum and the product of the roots of each equation.

(a) $x^2-7x+10=0$ [2]

(b) $2x^2+3x-5=0$ [2]

(c) $x^2+4x+4=0$ [2]

(d) $3x^2-2x-8=0$ [2]

22. For each cubic or quartic, state the sum and the product of the roots.

(a) $x^3-6x^2+11x-6=0$ [2]

(b) $2x^3-4x^2+x-3=0$ [3]

(c) $x^4-3x^3+2x-5=0$ [3]

23. Build the required equation.

(a) Write a quadratic with lead coefficient 1 whose roots sum to 5 and multiply to 6. [2]

(b) The roots of $x^2 + bx + 12 = 0$ are 3 and one other. Find the other root and b. [3]

(c) Find the value of k for which $x^2 - kx + 9 = 0$ has equal roots. [3]

(d) Write a cubic with lead coefficient 1 whose roots are -1, 2 and 4. [3]

24. The roots of $2x^2 - 7x + 6 = 0$ are α and β. Without solving the equation, find each value.

(a) $\alpha + \beta$ and $\alpha\beta$ [2]

(b) $\dfrac{1}{\alpha} + \dfrac{1}{\beta}$ [2]

(c) $\alpha^2 + \beta^2$ [2]

(d) $(\alpha - \beta)^2$ [3]

25. The roots of $x^2 + 5x + 3 = 0$ are α and β.

(a) Write down $\alpha + \beta$ and $\alpha\beta$. [2]

(b) Find the sum and product of $\alpha + 2$ and $\beta + 2$. [3]

(c) Hence find a quadratic equation with integer coefficients whose roots are $\alpha + 2$ and $\beta + 2$. [2]

26. The roots of $x^3 - 4x^2 + x + 6 = 0$ are α, β and γ.

(a) Write down $\alpha + \beta + \gamma$ and $\alpha\beta\gamma$. [2]

(b) Write down $\alpha\beta + \beta\gamma + \gamma\alpha$. [2]

(c) Hence find $\alpha^2 + \beta^2 + \gamma^2$. [3]

Chapter 7 · Polynomials

Reflections

IM The history behind this chapter is unusual. In Italy in the 1530s, mathematicians competed in public problem-solving contests, and a loser could lose his university post. Methods were therefore kept secret. Niccolò Tartaglia found a way to solve the cubic and told no one. Gerolamo Cardano persuaded him to share it under an oath of silence, then published it in 1545, together with his student Ferrari's solution of the quartic. Almost three hundred years passed before Niels Abel, who died of tuberculosis at twenty-six, proved that no such formula exists for the quintic. Évariste Galois, killed in a duel at twenty, explained exactly which equations can be solved by a formula and which cannot. To do this he created *group theory*, the mathematical study of symmetry, which is now used in particle physics and cryptography.

TOK Abel and Galois proved something surprising. The roots of a general quintic *exist*, but they cannot be written using only addition, subtraction, multiplication, division and roots. The equation $x^5 - x - 1 = 0$ has a real solution. That number is completely determined, and a computer can calculate it to a thousand decimal places. Even so, no formula of that kind can express it exactly. What does it mean, then, to "know" a number? Is such a root any less real than one you can write down?

RW Polynomials are used constantly in computing, because they need only multiplication and addition. Those are the two operations a processor performs fastest. The curved shapes in fonts and animation are drawn with *Bézier curves*, which are polynomial curves whose shape is controlled by a few chosen points. *Error-correcting codes* are built from polynomials as well. These are methods of adding extra information to a message so that it can still be read after part of it is damaged, which is how a scratched disc still plays and how a faint signal from a spacecraft is recovered. Polynomials are also used to fit smooth models to scattered measurements in science and finance.

EXT The factor theorem raises a question it cannot answer. How many roots does a polynomial really have? Over the real numbers, $x^2 + 1$ has none. The **Fundamental Theorem of Algebra** gives the answer. Over the *complex* numbers of Chapter 18, every polynomial of degree n has exactly n roots, counted with multiplicity. The sum and product formulas then hold for every polynomial, because the roots they describe always exist.

End-of-Chapter Problems

1. Consider $f(x) = x^3 - 2x^2 - 5x + 6$.

(a) Show that $x = 1$ is a root. [2]
(b) Factorise $f(x)$ completely. [3]
(c) Solve $f(x) = 0$. [1]

2. Consider $f(x) = 2x^3 + 3x^2 - 11x - 6$.

(a) Show that $(x - 2)$ is a factor. [2]
(b) Factorise $f(x)$ completely. [3]

3. When $x^3 + ax + b$ is divided by $(x - 1)$ the remainder is 4, and $(x + 1)$ is a factor. Find a and b. [5]

4. The polynomial $x^3 - 3x^2 + kx - 8$ has $(x - 2)$ as a factor. Find k. [3]

5. Sketch $y = (x + 1)(x - 1)(x - 3)$, showing all intercepts and the shape. [4]

6. A quadratic $x^2 + px + q$ has roots 2 and -5. Find p and q. [3]

7. Consider $f(x) = x^3 + 2x^2 - x - 2$.

(a) Factorise $f(x)$ completely. [3]
(b) Solve $f(x) = 0$. [1]

8. Solve $x^3 - 7x + 6 = 0$. [4]

9. The roots of $x^2 - 5x + 3 = 0$ are α and β. Find $\alpha^2 + \beta^2$. [4]

10. The polynomial $x^3 + ax^2 + bx - 2$ has $(x - 1)$ and $(x - 2)$ as factors. Find a and b. [5]

11. A cubic has roots 1, 2 and 3.

(a) Write it in factored and expanded form (leading coefficient 1). [3]
(b) State the sum and product of the roots and verify them from the expanded form. [2]

12. Consider $f(x) = x^4 - 5x^2 + 4$.

(a) Solve $f(x) = 0$. [3]

(b) State how many x-intercepts the graph has. [1]

13. Show that $x = 1$ is a root of $x^3 + 4x^2 + x - 6 = 0$, and find the other roots. [4]

14. The equation $x^2 - kx + 9 = 0$ has equal roots. Use the sum and product of roots to find k. [4]

15. The roots of $3x^2 - x - 2 = 0$ are α and β. Find $\dfrac{1}{\alpha} + \dfrac{1}{\beta}$. [3]

16. Consider $f(x) = 2x^3 - x^2 - 13x - 6$.

(a) Show that $(x + 2)$ is a factor. [2]
(b) Factorise $f(x)$ completely and solve $f(x) = 0$. [4]

17. The remainder when $x^3 + cx + d$ is divided by $(x - 1)$ is 6, and $(x - 2)$ is a factor. Find c and d. [5]

18. Solve $2x^3 + 3x^2 - 3x - 2 = 0$. [5]

19. The cubic $x^3 + px^2 + qx + r$ has roots -1, 2 and 4. Find p, q and r. [5]

20. The roots of $x^2 - 6x + 7 = 0$ are α and β. Find $\alpha^2 + \beta^2$ and $(\alpha - \beta)^2$. [5]

21. The polynomial $x^3 - kx + 2$ has $(x - 2)$ as a factor.

(a) Find k. [2]
(b) Find the other two roots, in exact form. [4]

22. Sketch $y = x^2(x - 3)$, describing the behaviour at each root. [4]

23. The quadratic with roots α and β is $x^2 - 4x + 1$. Find a quadratic (leading coefficient 1) whose roots are $\alpha + 1$ and $\beta + 1$. [5]

24. The polynomial $x^3 - 2x^2 + x - 2$ factorises as $(x - 2)(x^2 + 1)$.

(a) Verify this by expansion. [2]
(b) State the only real root, and explain why there are no others. [2]

25. When a polynomial $p(x)$ is divided by $(x - 1)$ the remainder is 3, and by $(x - 2)$ the remainder is 5. Find the remainder when $p(x)$ is divided by $(x - 1)(x - 2)$. [6]

26. The roots of $x^3 - 6x^2 + 11x - 6 = 0$ are α, β, γ. Find $\alpha^2 + \beta^2 + \gamma^2$ without solving the equation. [5]

27. Prove that if a polynomial with integer coefficients has a rational root $\frac{p}{q}$ in lowest terms, then p divides the constant term and q divides the leading coefficient. [6]

28. The cubic $x^3 + px + q = 0$ has a repeated root. Show that $4p^3 + 27q^2 = 0$. [6]

29. Find all real values of k for which $x^3 - 3x + k = 0$ has three distinct real roots. [6]

30. Consider $f(x) = 2x^3 - 3x^2 - 11x + 6$.

(a) Show that $(x - 3)$ is a factor of $f(x)$. [2]
(b) Factorise $f(x)$ completely. [4]
(c) Solve $f(x) = 0$. [2]
(d) State the y-intercept, and describe the behaviour of $f(x)$ as $x \to \pm\infty$. [3]
(e) Sketch $y = f(x)$, showing all intercepts. [2]
(f) Hence solve the inequality $f(x) \leq 0$. [3]

31. The polynomial $p(x) = x^3 + ax^2 + bx + 6$ leaves a remainder of 12 when divided by $(x - 2)$, and $(x + 1)$ is a factor.

(a) Show that $4a + 2b = -2$ and $a - b = -5$. [4]
(b) Hence find a and b. [2]
(c) Factorise $p(x)$ as far as possible over the real numbers. [4]
(d) State how many real roots $p(x) = 0$ has, giving a reason. [2]
(e) Find the remainder when $p(x)$ is divided by $(x - 1)(x + 1)$. *(Hint: the remainder has the form $rx + s$)* [4]

32. The cubic equation $x^3 + px^2 + qx + r = 0$ has roots α, β and γ.

(a) Write down $\alpha + \beta + \gamma$, $\alpha\beta + \beta\gamma + \gamma\alpha$ and $\alpha\beta\gamma$ in terms of p, q and r. [3]
(b) The equation $x^3 - 6x^2 + 3x + 10 = 0$ has roots α, β, γ. Find $\alpha^2 + \beta^2 + \gamma^2$ without solving the equation. [4]
(c) Find $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$. [3]
(d) Given that one root is -1, find the other two. [3]
(e) Write down a cubic equation with integer coefficients whose roots are 2α, 2β and 2γ. [3]

Challenge Questions

33. **Power sums of the roots.** The equation $x^2 - px + q = 0$ has roots α and β. For each integer $k \geq 0$ define

$$S_k = \alpha^k + \beta^k.$$

(a) Write down the values of S_0 and S_1. [2]
(b) Show that $S_2 = p^2 - 2q$. [2]
(c) Find S_3 in terms of p and q. [3]
(d) Explain why $\alpha^k = p\,\alpha^{k-1} - q\,\alpha^{k-2}$ for every $k \geq 2$, and deduce that $S_k = p\,S_{k-1} - q\,S_{k-2}$. [4]
(e) The equation $x^2 - 3x + 1 = 0$ has roots α and β. Use the recurrence to find S_2, S_3 and S_4. [4]
(f) The roots of $x^2 - 3x + 1 = 0$ are irrational. Explain why S_k is nevertheless an integer for every k. [3]

34. Palindromic polynomials. A polynomial is *palindromic* if its list of coefficients reads the same forwards as backwards. For example $x^4 + x^3 - 4x^2 + x + 1$ has coefficients $1, 1, -4, 1, 1$.

(a) Explain why $x = 0$ is never a root of a palindromic polynomial. [2]
(b) Let $P(x)$ be palindromic of degree n. Show that $x^n P\left(\frac{1}{x}\right) = P(x)$, and deduce that if α is a root of P then so is $\frac{1}{\alpha}$. [4]
(c) Divide $x^4 + x^3 - 4x^2 + x + 1 = 0$ through by x^2 and substitute $u = x + \frac{1}{x}$. Show that the equation becomes $u^2 + u - 6 = 0$. [4]
(d) Hence find all four roots of $x^4 + x^3 - 4x^2 + x + 1 = 0$ in exact form. [5]
(e) Show that every palindromic polynomial of odd degree has $x = -1$ as a root. [3]